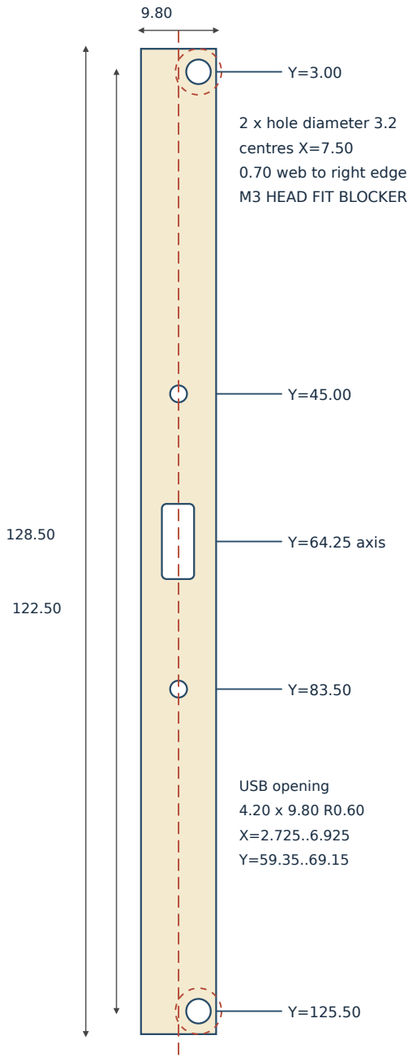
BIG MODULAR | 2HP NOMINAL ASSEMBLY | A1 WIP

INCOMPLETE / NON-OPERATIONAL. Drawing is a fit-test proposal, NOT fabrication or mechanical qualification.

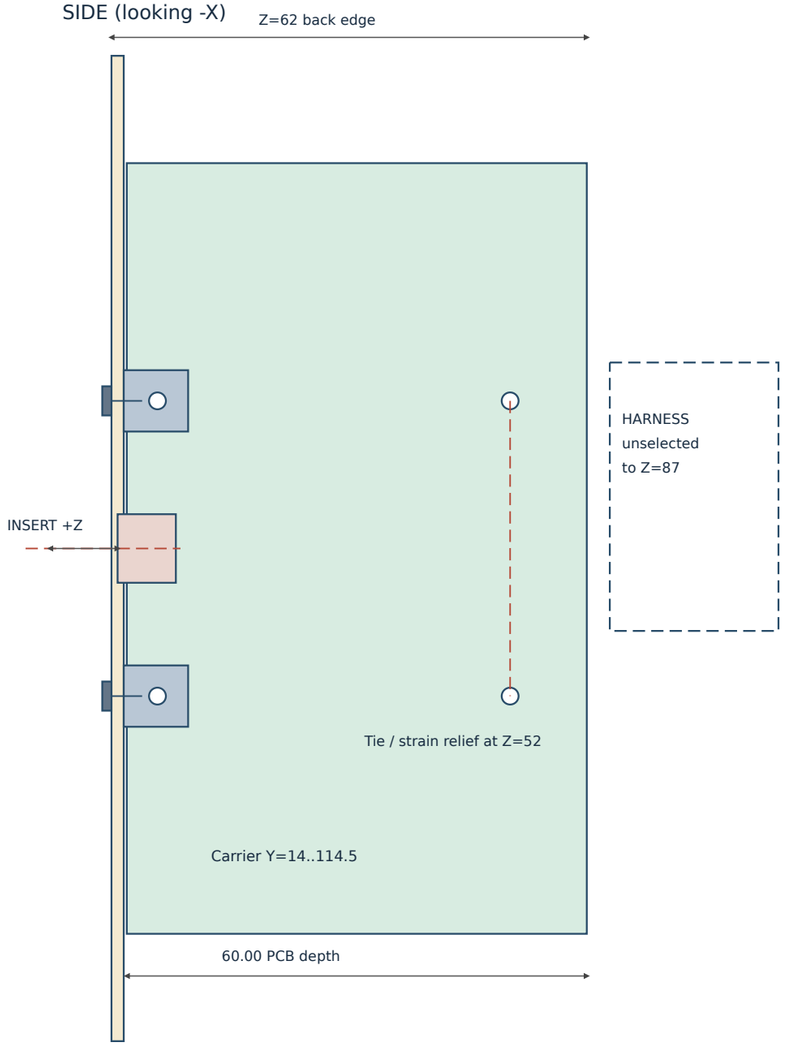
All dimensions mm. Panel origin: upper left front; X right, Y down, Z inward. Dimensions govern; do not scale prints.

FRONT (looking +Z)



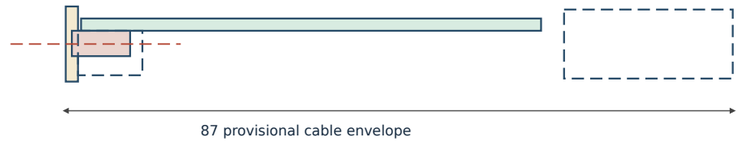
Panel: FR4 1.60 (not Doepfer 2 mm Al).
A 6 mm M3 head overhangs right by 0.7.
Ø2.2 bracket holes X=4.9, Y=45 / 83.5.

SIDE (looking -X)



Carrier: 100.5 x 60 x 1.6; plane YZ.
Front support holes (Y,Z): (45,6), (83.5,6).
Rear holes (45,52), (83.5,52): tie only; not rail mounts.

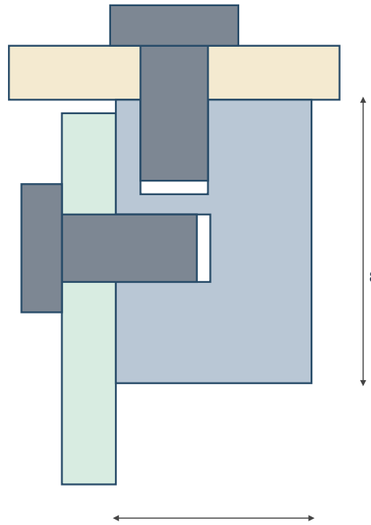
TOP (looking +Y)



Panel Z=0..1.6; carrier front Z=2 (0.4 rear gap).
PCB bottom X=1.57; component face X=3.17.
USB mouth (X,Y,Z)=(4.90,64.25,0.80).
Plug axis +Z; plug shell fit / overmould NOT qualified.

2 x RIGHT-ANGLE INTERPOSER BLOCKS

Solid machined aluminium; no bent/soldered FR4 brackets.



Each block: X 5.80 x Y 8.00 x Z 8.40.
Origin at (3.17,Yc-4,1.6), Yc=45 / 83.5.
Panel hole: Xlocal=1.73, Ylocal=4, axis Z.
Carrier hole: Ylocal=4, Zlocal=4.40, axis X.
Both M2 x 0.4 blind taps.
Full thread >=2.6; TOTAL depth <=2.8,
including drill point. Flat-bottom process.
4 x M2 x 4; engagement nominal 2.4.
Side head h<=1.2, diameter <=3.8.
Exact low-head screw is UNSELECTED.
Faces/axes target +/-0.05; block +/-0.10.
Bare FR4 contact; copper keepout required.
Two support points, Y separation 38.50.

Panel/PCB thickness tolerance changes engagement and alignment.
Screw heads remain inside X=0.37..8.97 nominal (2HP width 9.8).
Keep Y<10 and Y>118.5 free for rail access; PCB begins at Y=14.
No independent USB shell cradle supplied: stakes -> PCB -> blocks.
Insertion force / cable torque / FR4 flex MUST be tested.

MOUNTING DECISION: Doepfer A100FP.gif shows x=7.5, d=3.2, y=3 from both edges; pitch 122.5.

CORRECTED rail holes to x=7.5, diameter 3.2: 0.7 FR4 web. A 5 x 3.2 slot runs off-panel, so NOT used.

Standard 6 mm M3 heads overhang by 0.7: adjacent-module conflict. Narrow head / panel strength NOT resolved.

M3 x 6 through 1.6 panel + 0.5 washer leaves 3.9 nominal thread reach: verify actual rail depth; do not bottom out.

Assembly: tap blocks -> bolt to bare carrier -> solder J1 stakes -> loosely bolt panel -> align USB aperture -> tighten -> fit tie/harness -> rack.

Profile/aperture target +/-0.10; only 0.43/0.445 nominal shell clearance per side. Trial plugs / worst-case stack remain blockers.

Sources: doepfer.de/a100_man/A100FP.gif + a100m_e.htm; GCT USB4105 drawing geometry retained from A0. See mechanics.md.